Mohammad Asgar Khan Dalwai

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EDUCATION

Master of Science in Data Science May 2027
Arizona state university, Tempe AZ GPA 4.0
Relevant Courses: Analyzing Big Data, Statistical Machine Learning, Statistics for Data Analysts, Data Processing at Scale, Data Mining, Data visualization, Advanced Database Management System, Advanced Computer Network Security

Bachelor of Technology in computer Science and Business System June 2024
Bharati Vidyapeeth (DU) DET, Mumbai India GPA 3.4
Relevant Courses: Data Structures and Algorithms, Computer Networks, Database Management Systems, Machine Learning, Internet of Things, Operating Systems, Information Security

SKILLS

Programming Languages: Python, SQL, Java

Data Science Skills: Pandas, NumPy, Scikit Learn, Matplotlib, MobileNetV2, Tableau, MySQL, Jupyter notebook, Clustering, NLP, OpenCV, TKinter, XGBoost, SHAP

Tools: PyCharm, MySQL Workbench, VScode, Git.

Databases:MySQL, PostgreSQL

Certifications: Industrial Project Completion Certificate, Foundational course in programming and data science

PROFESSIONAL EXPERIENCE

Artificial Intelligence Trainee Feb 2022 – April 2022 Mumbai, India
Languify

- Built an Automatic Speech Recognition (ASR) system combining acoustic and language modeling to recognize Indian-accented English speech.
- Implemented MFCC feature extraction, HMM decomposition, and ANN-based deep learning to integrate acoustic modeling, language modeling, and data preprocessing, enhancing overall transcription accuracy.
- Reduced WER to ~18% using custom dataset consisting of 500+ audio recordings by training and evaluating deep learning models on this data, which increased accuracy in speech recognition by ~15%.

ACADEMIC PROJECTS

Multiple Eye Disease Detection Using machine learning Fall 2023
Bharati Vidyapeeth (DU) DET

- Designed and implemented a deep learning system to detect cataract, glaucoma, and diabetic retinopathy from retinal images.
- Utilized MobileNetV2 CNN architecture for efficient feature extraction and classification, achieving high diagnostic accuracy.
- Employed image preprocessing, data augmentation, and MFCC- based feature analysis to improve model performance and generalization.
- Built a Flask based web interface for real-time image upload and prediction, enabling accessible, automated eye disease screening.

Loan Approval Prediction Fall 2025
Arizona State University

- Built a machine learning pipeline to support loan approval and risk assessment, helping identify high-risk applications and improve consistency in lending decisions.
- Compared multiple models and analyzed feature importance to surface the factors most influencing approval risk, improving interpretability for business and risk teams.
- Engineered key risk features and improved the model's ability to detect applicants likely to default.
- Contributed to a predictive system that can help lenders reduce default exposure, prioritize manual reviews, and streamline approval workflows.

PUBLICATIONS

Automated Attendance Management System Using Facial Recognition
IAES International Journal of Artificial Intelligence (IJ-AI), Vol. 13, No 1, March 2024 Part II

Multiple Eye disease Detection Using Machine learning
International Conference on Intelligent Computing and Communication Techniques (ICICCT-2024), held from 28th to 29th June, 2024